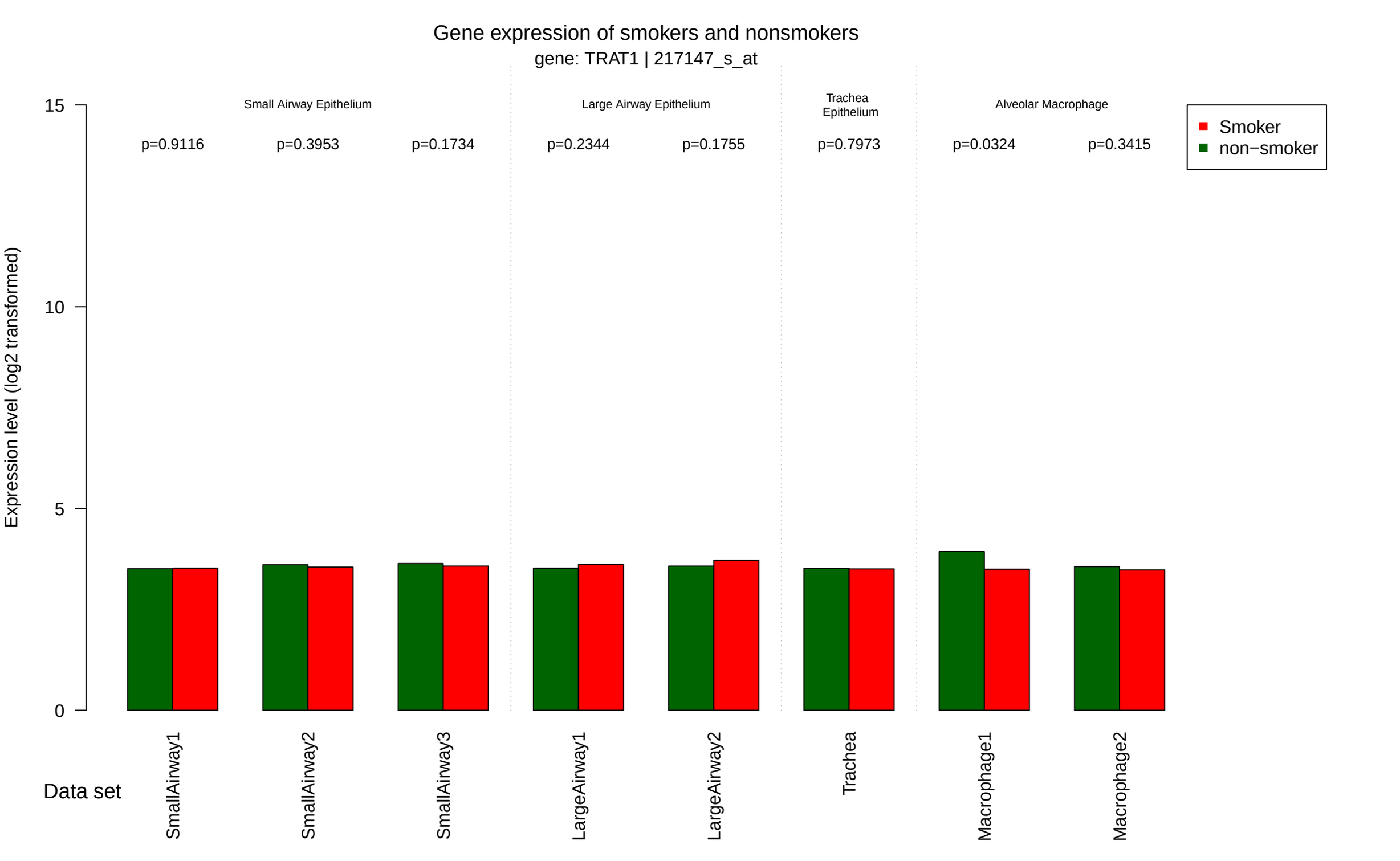
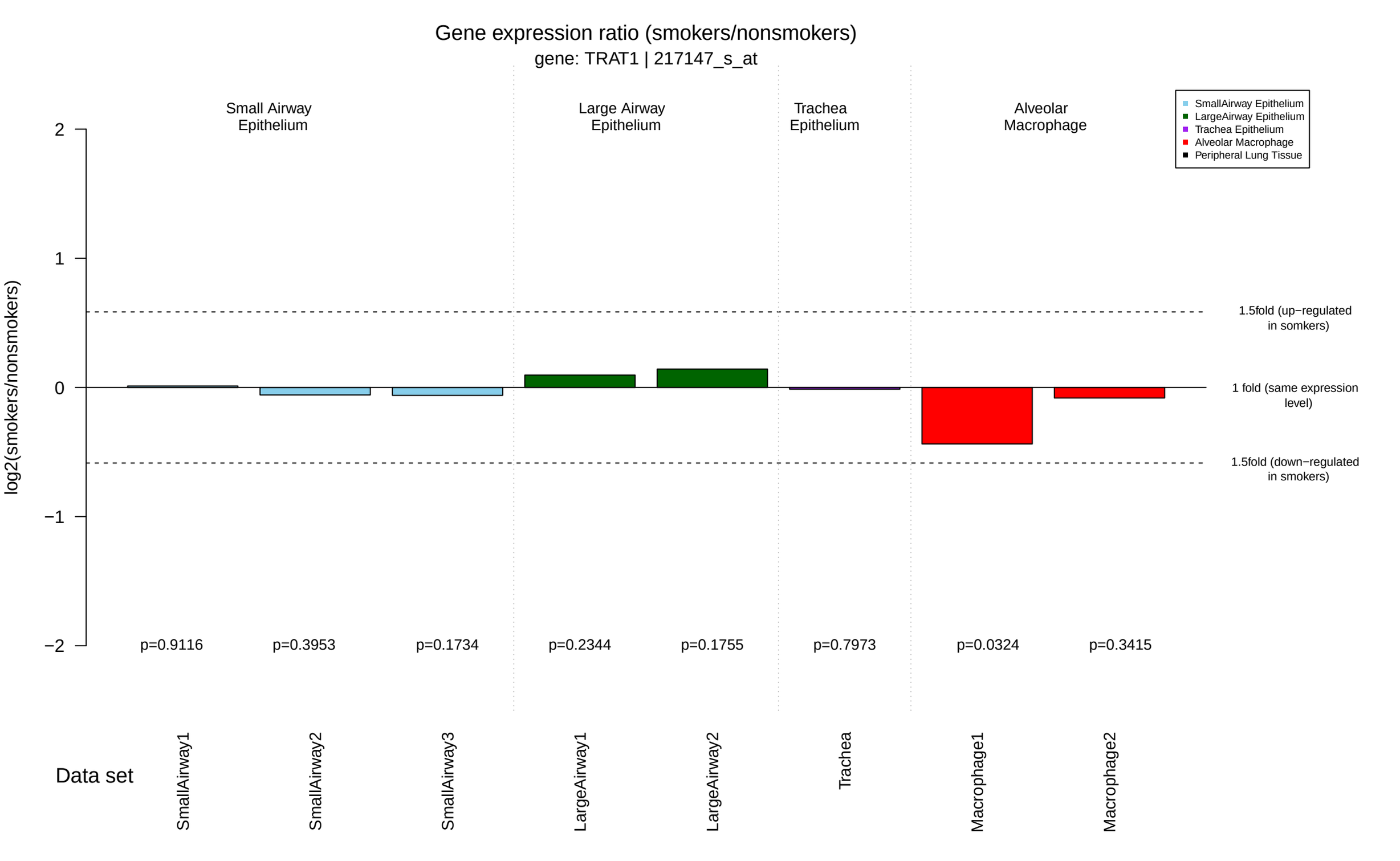
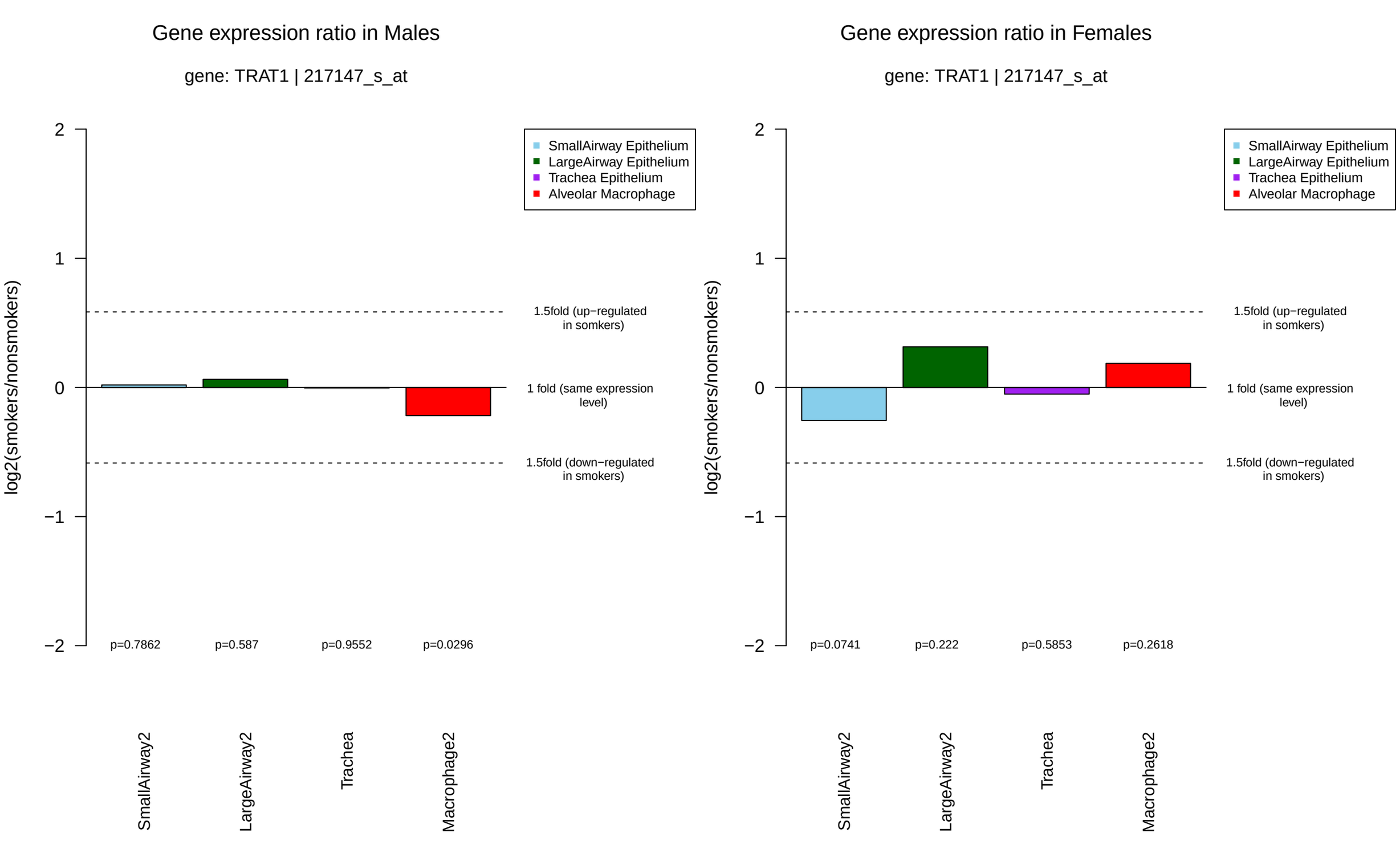
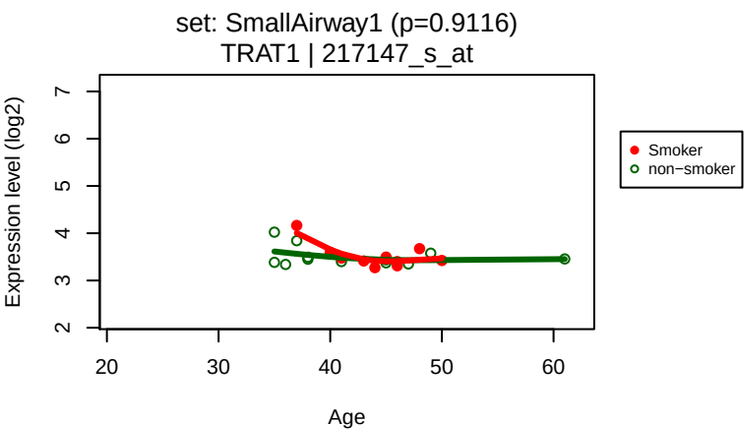
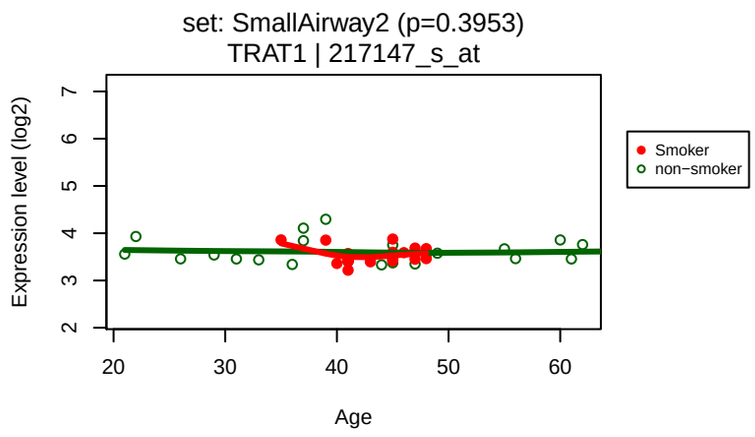
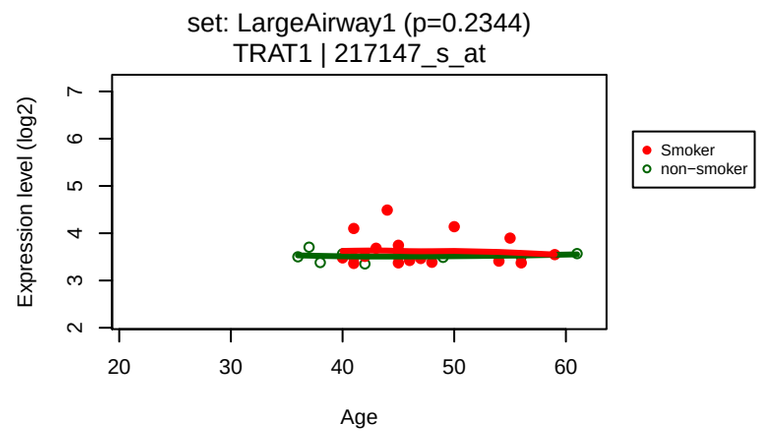

Gene expression of smokers and nonsmokers
gene: TRAT1 | 217147_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9116
p=0.3953
p=0.1734
p=0.2344
p=0.1755
p=0.7973
p=0.0324
p=0.3415
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

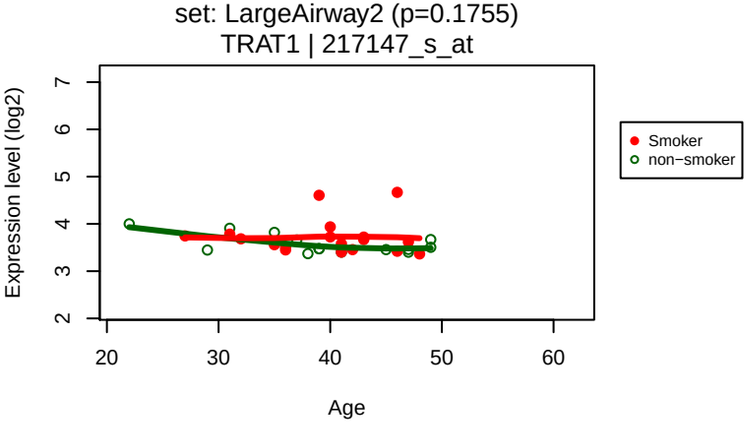

Gene expression ratio (smokers/nonsmokers)
gene: TRAT1 | 217147_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9116
p=0.3953
p=0.1734
p=0.2344
p=0.1755
p=0.7973
p=0.0324
p=0.3415
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

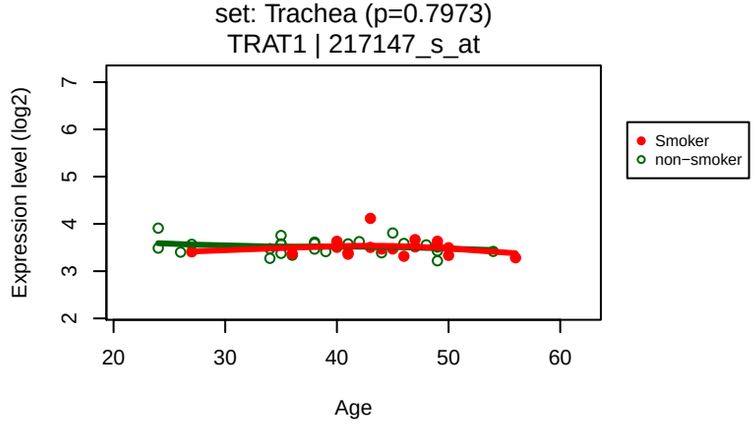

Gene expression ratio in Males
gene: TRAT1 | 217147_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7862
p=0.587
p=0.9552
p=0.0296
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TRAT1 | 217147_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0741
p=0.222
p=0.5853
p=0.2618
SmallAirway2
LargeAirway2
Trachea
Macrophage2

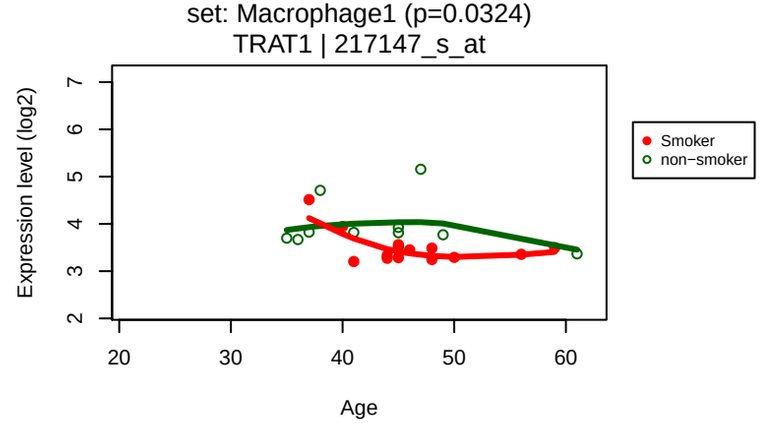

set: SmallAirway1 (p=0.9116)
TRAT1 | 217147_s_at
Expression level (log2)
Age
Smoker
non−smoker

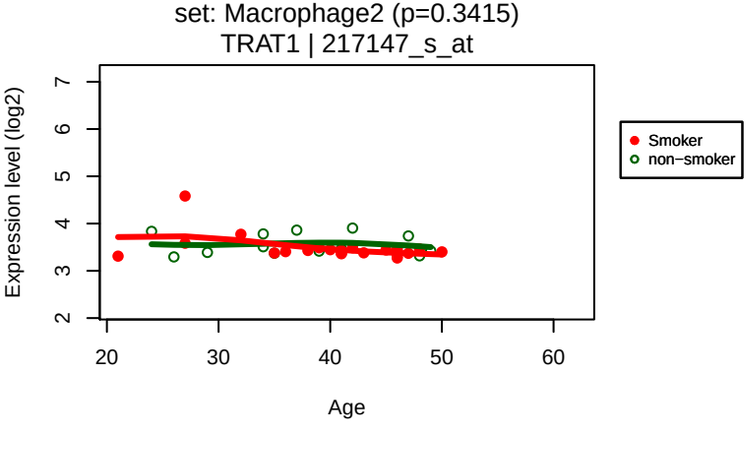

set: SmallAirway2 (p=0.3953)
TRAT1 | 217147_s_at
Expression level (log2)
Age
Smoker
non−smoker

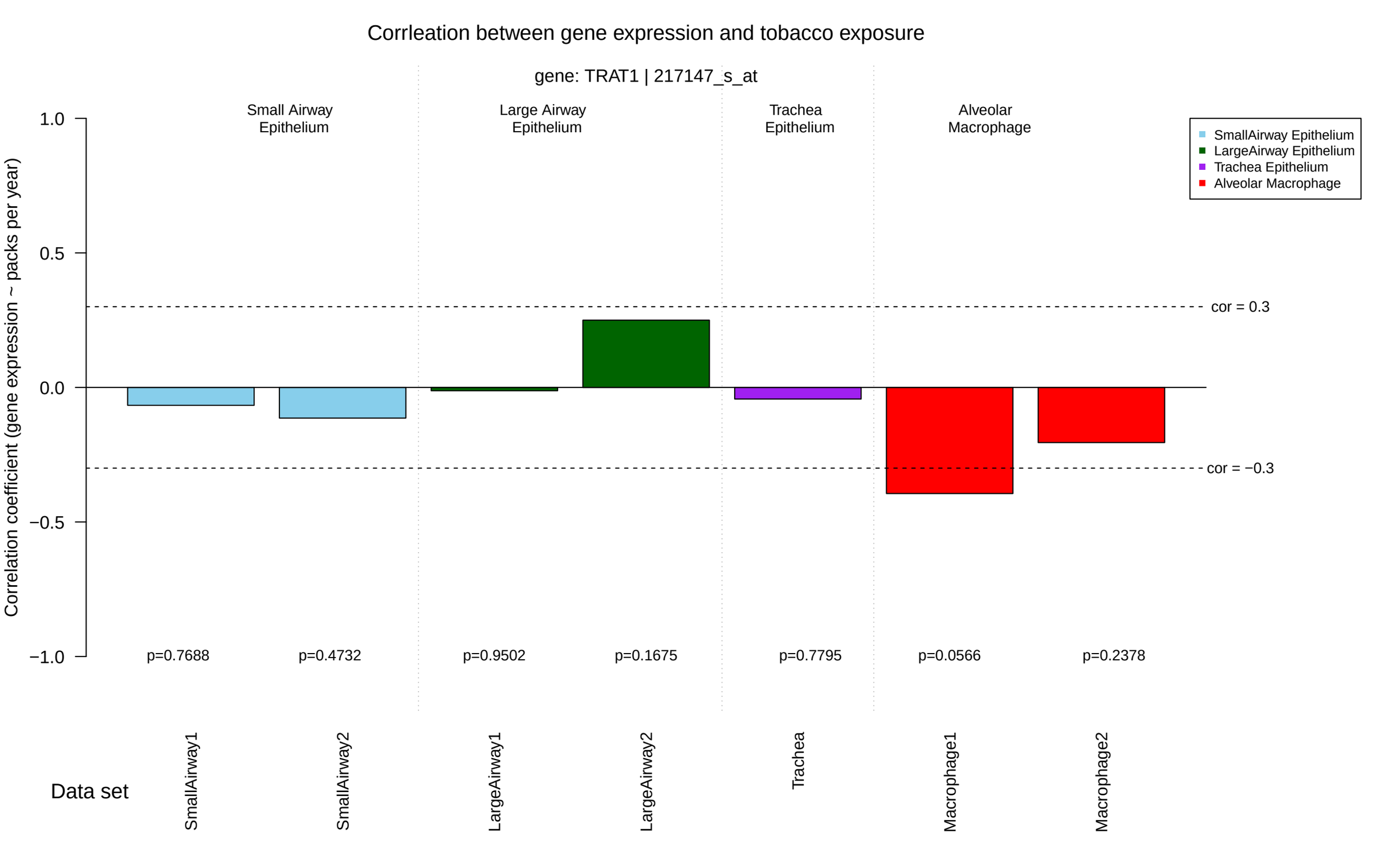

set: LargeAirway1 (p=0.2344)
TRAT1 | 217147_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1755)
TRAT1 | 217147_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7973)
TRAT1 | 217147_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0324)
TRAT1 | 217147_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3415)
TRAT1 | 217147_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TRAT1 | 217147_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7688
p=0.4732
p=0.9502
p=0.1675
p=0.7795
p=0.0566
p=0.2378
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2